

SCRIPT-4.R

Tory Ferrer

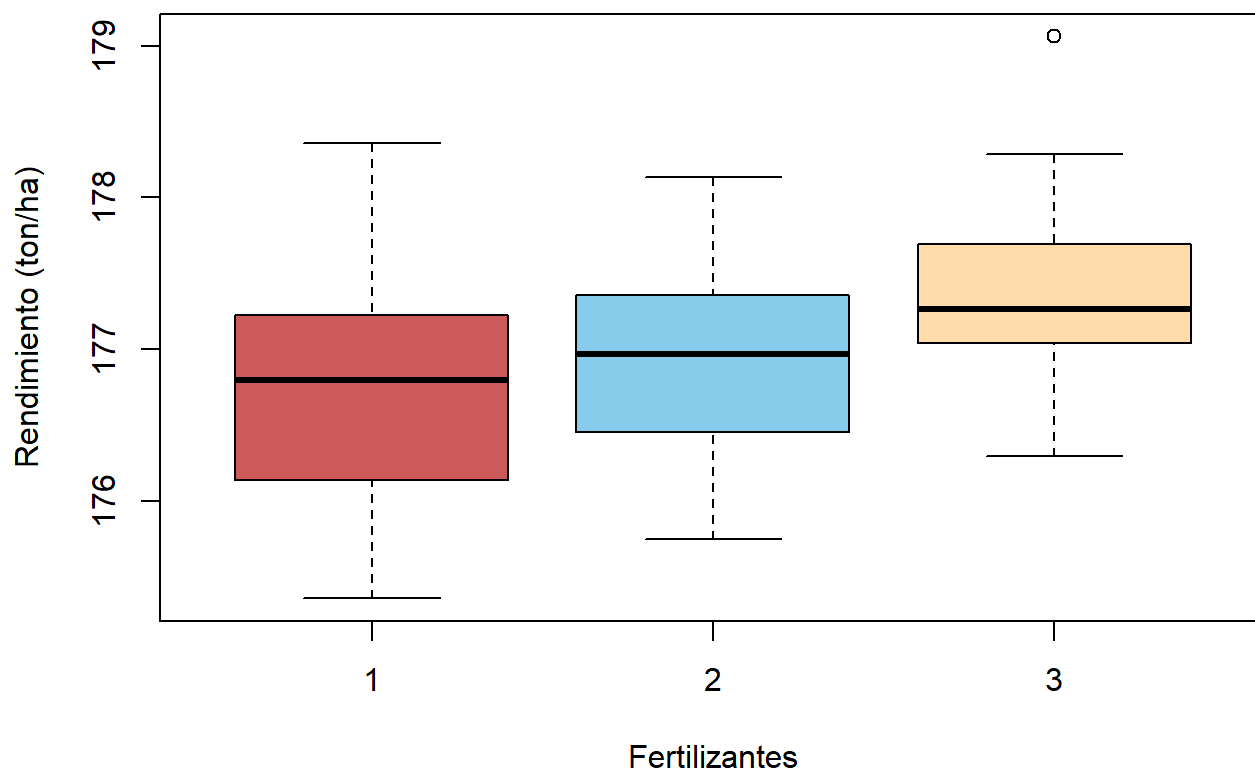
2025-09-18

```
# =====  
# Análisis de varianza  
# Productividad de cultivos  
# 18/09/2025  
# LUIS MIGUEL TORIBIO FERRER  
# SEMANA 7  
# =====
```

```
url <- "https://raw.githubusercontent.com/mgtagle/Posgrado_Estadistica_2025/refs/heads/main/crop.data.csv"  
crop <- read.csv(url, header = T)  
crop$density <- as.factor(crop$density)  
crop$block <- as.factor(crop$block)  
crop$fertilizer <- as.factor(crop$fertilizer)  
summary(crop)
```

```
## density block fertilizer yield  
## 1:48 1:24 1:32 Min. :175.4  
## 2:48 2:24 2:32 1st Qu.:176.5  
## 3:24 3:32 Median :177.1  
## 4:24 Mean :177.0  
## 3rd Qu.:177.4  
## Max. :179.1
```

```
colores <- c("indianred", "skyblue", "navajowhite")  
boxplot(crop$yield ~ crop$fertilizer,  
        col = colores,  
        xlab = "Fertilizantes",  
        ylab = "Rendimiento (ton/ha)")
```



```
library(vioplplot)
```

```
## Warning: package 'vioplplot' was built under R version 4.5.1
```

```
## Cargando paquete requerido: sm
```

```
## Warning: package 'sm' was built under R version 4.5.1
```

```
## Package 'sm', version 2.2-6.0: type help(sm) for summary information
```

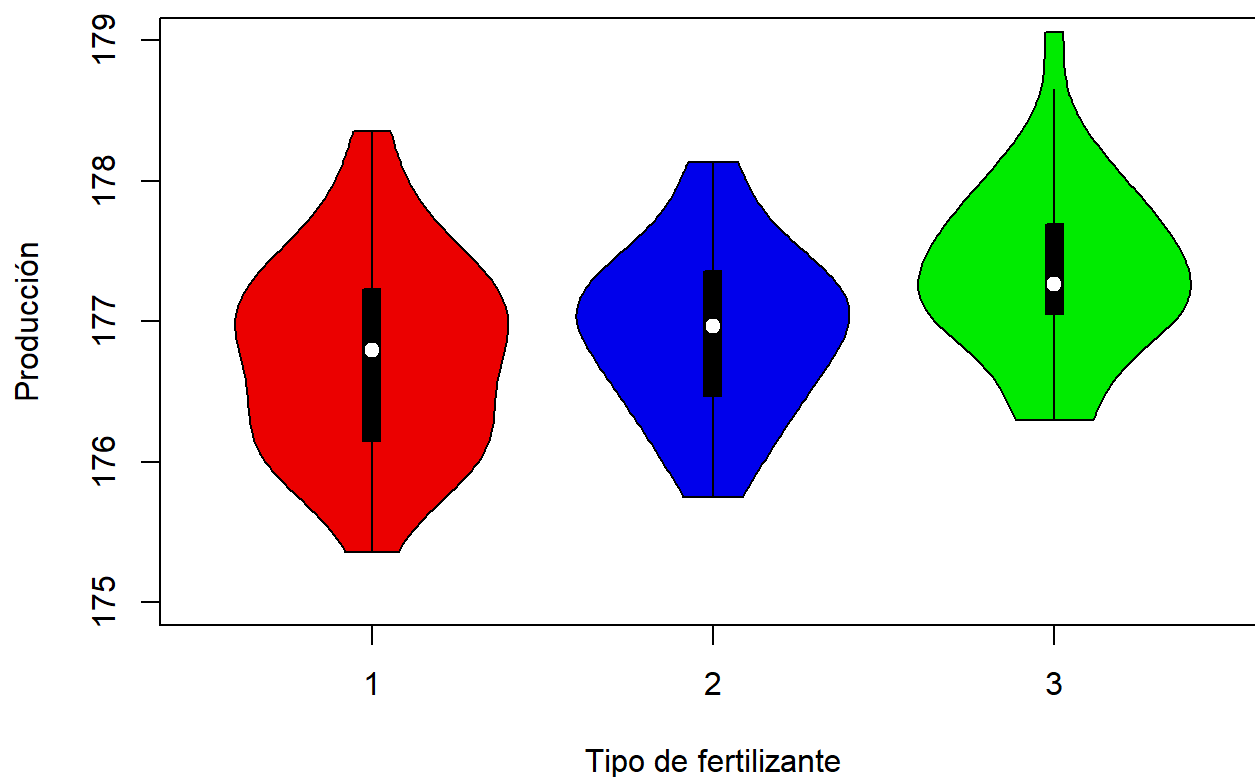
```
## Cargando paquete requerido: zoo
```

```
##  
## Adjuntando el paquete: 'zoo'
```

```
## The following objects are masked from 'package:base':  
##  
## as.Date, as.Date.numeric
```

```
vioplot(crop$yield ~ crop$fertilizer,
        main = "Producción por fertilizante",
        col= c("red2","blue2", "green2"),
        ylab = "Producción",
        xlab = "Tipo de fertilizante",
        ylim = c(175, 179))
```

Producción por fertilizante



```
tapply(crop$yield, crop$fertilizer, mean)
```

```
##      1      2      3
## 176.7570 176.9332 177.3562
```

```
tapply(crop$yield, crop$fertilizer, var)
```

```
##      1      2      3
## 0.4691199 0.3295526 0.3589464
```

```
# aplicar prueba de normalidad de datos para cada
# fertilizante combinando subset y shapiro

shapiro.test(subset(crop$yield, crop$fertilizer == "1"))
```

```
##
## Shapiro-Wilk normality test
##
## data: subset(crop$yield, crop$fertilizer == "1")
## W = 0.97914, p-value = 0.7743
```

```
shapiro.test(subset(crop$yield, crop$fertilizer == "2"))
```

```
##
## Shapiro-Wilk normality test
##
## data: subset(crop$yield, crop$fertilizer == "2")
## W = 0.98329, p-value = 0.8875
```

```
shapiro.test(subset(crop$yield, crop$fertilizer == "3"))
```

```
##
## Shapiro-Wilk normality test
##
## data: subset(crop$yield, crop$fertilizer == "3")
## W = 0.95878, p-value = 0.2542
```

```
# Prueba de Bartlett
bartlett.test(crop$yield ~ crop$fertilizer)
```

```
##
## Bartlett test of homogeneity of variances
##
## data: crop$yield by crop$fertilizer
## Bartlett's K-squared = 1.0622, df = 2, p-value = 0.5879
```

```
# Prueba de ANOVA

crop.aov <- aov(crop$yield ~ crop$fertilizer)
summary(crop.aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## crop$fertilizer  2   6.07   3.0340    7.863  7e-04 ***
## Residuals      93  35.89   0.3859
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# LSD determinar el valor
```

```
qt(0.975, 93)
```

```
## [1] 1.985802
```

```
sqrt((2*0.3859)/32) * qt(0.975, 93)
```

```
## [1] 0.3083992
```

```
tapply(crop$yield, crop$fertilizer, mean)
```

```
##          1          2          3  
## 176.7570 176.9332 177.3562
```

```
# primer diferencia de medias F1 vs F2  
176.757 - 176.9332
```

```
## [1] -0.1762
```

```
# primer diferencia de medias F2 vs F3  
176.9332 - 177.3562
```

```
## [1] -0.423
```

```
# # primer diferencia de medias F1 vs F3  
176.757 - 177.3562
```

```
## [1] -0.5992
```

```
# Prueba de Tukey
```

```
sqrt((0.3859)/32) * qtukey(0.95, nmeans = 3, df = 93)
```

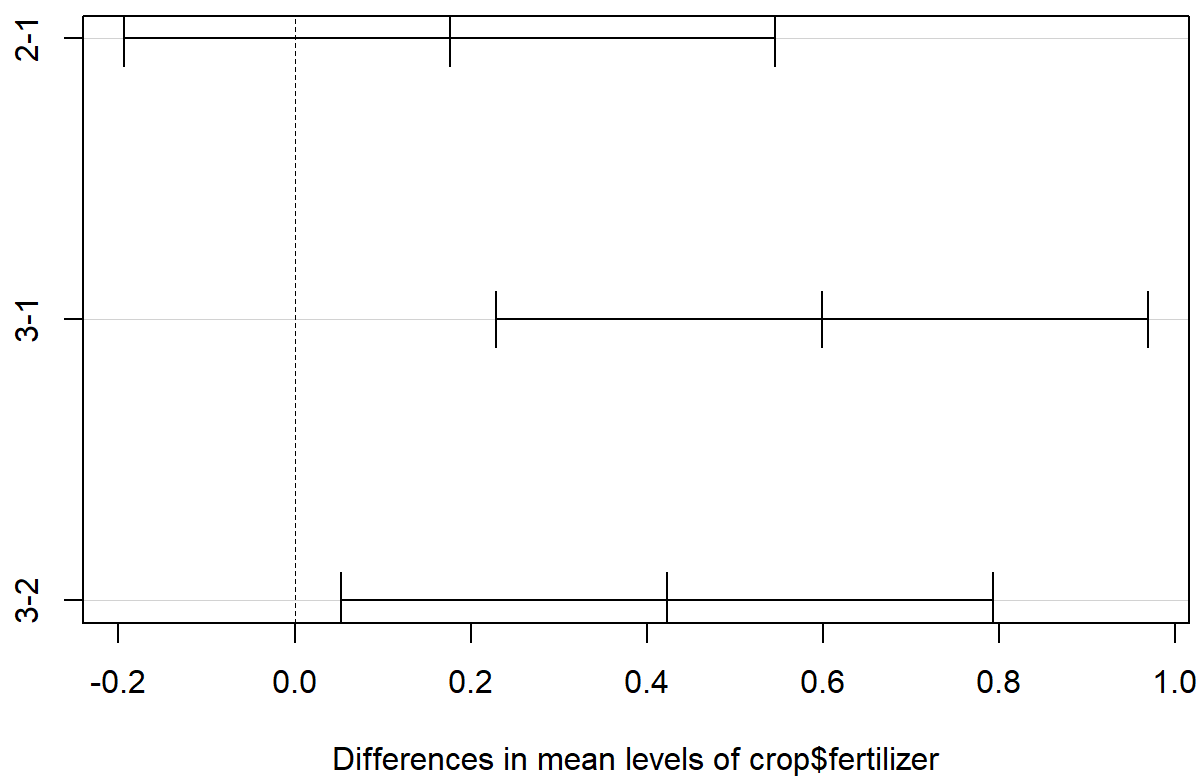
```
## [1] 0.3699006
```

```
TukeyHSD(crop.aov, )
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = crop$yield ~ crop$fertilizer)
##
## $`crop$fertilizer`
##      diff      lwr      upr    p adj
## 2-1 0.1761687 -0.19371896 0.5460564 0.4954705
## 3-1 0.5991256  0.22923789 0.9690133 0.0006125
## 3-2 0.4229569  0.05306916 0.7928445 0.0208735
```

```
plot(TukeyHSD(crop.aov))
```

95% family-wise confidence level



```
library(ggplot2)

ggplot(crop, aes(x=fertilizer, y = yield, fill = fertilizer))+
  geom_violin(alpha = 0.5)+
  geom_jitter(col = "indianred")+
  geom_boxplot(width = 0.1, col = "white", alpha = 0.5)+
  theme_light()+
  labs(x = "Fertilizante",
       y = "Rerndimiento (ton/ha)")
```

